

Activation of NF- κ B is required for hypertrophic growth of primary rat neonatal ventricular cardiomyocytes

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The transcription factor NF- κ B regulates expression of genes that are involved in inflammation, immune response, viral infection, cell survival, and division. However, the role of NF- κ B in hypertrophic growth of terminally differentiated cardiomyocytes is unknown. Here we report that NF- κ B activation is required for hypertrophic growth of cardiomyocytes. In cultured rat primary neonatal ventricular cardiomyocytes, the nuclear translocation of NF- κ B and its transcriptional activity were stimulated by several hypertrophic agonists, including phenylephrine, endothelin-1, and angiotensin II. The activation of NF- κ B was inhibited by expression of a "supersuppressor" I κ B α mutant that is resistant to stimulation-induced degradation and a dominant negative I κ B kinase (IKK β) mutant that can no longer be activated by phosphorylation. Furthermore, treatment with phenylephrine induced I κ B α degradation in an IKK-dependent manner, suggesting that NF- κ B is a downstream target of the hypertrophic agonists. Importantly, expression of the supersuppressor I κ B α mutant or the dominant negative IKK β mutant blocked the hypertrophic agonist-induced expression of the embryonic gene atrial natriuretic factor and enlargement of cardiomyocytes. Conversely, overexpression of NF- κ B itself induced atrial natriuretic factor expression and cardiomyocyte enlargement. These findings suggest that NF- κ B plays a critical role in the hypertrophic growth of cardiomyocytes and may serve as a potential target for the intervention of heart disease.

The transcription factor NF- κ B regulates expression of many genes that are involved in inflammation, immune response, viral infection, and cell survival (1–5). In most resting cells, NF- κ B is bound to its cytoplasmic inhibitory proteins, I κ B (α , β , and ϵ), and remains in the cytoplasm as a latent form transcription factor (1). Upon stimulation, the I κ B kinase (IKK) complex (6–12), which is composed of two catalytic subunits IKK α and IKK β and a regulatory subunit IKK γ (12–14), is activated and it in turn phosphorylates I κ B proteins on specific Ser residues (Ser-32 and -36 on I κ B α and Ser-19 and -23 on I κ B β) (15–17). The phosphorylation triggers ubiquitination-dependent degradation of I κ B proteins by the 26S proteasome, resulting in the release of NF- κ B (4, 18, 19). Subsequently, NF- κ B translocates into the nucleus, where it stimulates transcription of specific target genes (4).

NF- κ B also plays a role in regulating cell growth. Genetic disruption of members of the NF- κ B family, such as p65, p50, or c-Rel, impairs proliferation of lymphocytes (20–22). Furthermore, NF- κ B can be activated by oncogenic Ras and Raf and is involved in Ras-induced transformation of NIH 3T3 or liver epithelial cells (23, 24). Recent studies also show that NF- κ B regulates expression of cyclin D1 and its activation is required for the G₁-S transition (25, 26). The role of NF- κ B in hypertrophic growth of terminally differentiated cells, however, has yet to be vigorously studied.

Cardiomyocytes are terminally differentiated cells (27). In response to various extracellular stimuli, cardiomyocytes grow in

a hypertrophic manner, an event that is characterized by enlargement of individual cell size, an increase in the content of contractile proteins such as myosin heavy chain, and expression of embryonic genes such as atrial natriuretic factor (ANF) (27, 28). The collective result is cardiac hypertrophy, which is an adaptive and compensatory response in nature (28). However, sustained cardiac hypertrophy can set overt heart failure in motion (27).

In this report, we show that in rat primary neonatal ventricular cardiomyocytes, NF- κ B activity was stimulated by several hypertrophic agonists in an IKK-dependent manner. More importantly, hypertrophic agonist-induced expression of the embryonic gene ANF and enlargement of cardiomyocytes were abrogated by inhibition of NF- κ B activity but induced by overexpression of members of the NF- κ B family. These results demonstrate that the NF- κ B signaling pathway is required for hypertrophic growth of cardiomyocytes, a process that may lead to heart failure (27, 29).

Materials and Methods

Cell Culture. Rat neonatal ventricular cardiomyocytes were prepared from hearts of 2- to 3-day-old Sprague–Dawley rats as described (30). After separation from nonmyocyte fibroblasts through preplating, myocytes were purified by Percoll gradient, replated at a density of 1.5×10^5 in 12-well plates precoated with 1% gelatin, and grown in plating media (DMEM and Media 199, 4:1), supplemented with 2 mM glutamine, 100 units/ml penicillin, 100 mg/ml streptomycin, and 0.1 mmol/liter BrdUrd. The inclusion of BrdUrd resulted in inhibition of the growth of cardiac fibroblasts.

cDNA Constructs. Expression vectors encoding hemagglutinin (HA)-tagged IKK β , the dominant negative HA-IKK β [177Ala/181Ala] mutant, in which Ser-177 and Ser-181 were replaced by nonphosphorylatable alanine (Ala), HA-I κ B α , the "supersuppressor" HA-I κ B α (32Ala/36Ala) mutant, in which Ser-32 and Ser-36 were replaced by Ala, HA-p65, and c-Rel, members of the NF- κ B family, have been described. Reporter genes of ANF-luciferase (ANF-Luc), in which the full-length rat ANF promoter (–3,003) was fused to the firefly Luc gene, and 2 $\times$ NF- κ B Luc (NF- κ B-Luc), in which two copies of the κ B promoter containing the NF- κ B-binding site was fused to Luc, also have been described (30, 31). Recombinant adenoviral vector (Ad) encoding the dominant negative I κ B α (32Ala/36Ala)

Abbreviations: HA, hemagglutinin; PE, phenylephrine; ET-1, endothelin-1; AngII, angiotensin II; ANF, atrial natriuretic factor; GFP, green fluorescent protein; TNF, tumor necrosis factor; Ad, adenoviral vector; Luc, luciferase; IKK, I κ B kinase.

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mutant was a gift from David A. Brenner (University of North Carolina, Chapel Hill) (32), and Ad/green fluorescent protein (GFP) (33) was a gift from Jialing Xiang (University of Chicago, Chicago).

Fusion Protein. Glutathione *S*-transferase-I κ B α was purified by glutathione-agarose affinity chromatography, as described (7, 31).

Transfections, Viral Infection, and Luc Assays. Primary neonatal cardiomyocytes were transiently transfected with the 2 $\times$ NF- κ B-Luc reporter gene or the ANF-Luc reporter gene with or without various expression vectors encoding HA-IKK β , the dominant negative mutant HA-IKK β (177Ala/181Ala), the supersuppressor HA-I κ B α (32Ala/36Ala) mutant, HA-p65, or c-Rel, by using Tfx-20 (Promega) according to the manufacturer's procedure. For Ad-mediated gene delivery, cells were infected with Ad/I κ B α (32Ala/36Ala) or Ad/GFP at 100 multiplicity of infection each, as described (33). After transfection or infection (12–16 h), cells were washed three times and maintained in serum-free medium with or without various cardiac hypertrophic agonists, as indicated in the figure legends. Luc assays were performed as described (31).

Protein Kinase Assays and Immunoblotting Analysis. IKK β assays were performed as described (31). Primary cardiomyocytes were transfected with the expression vector encoding HA-IKK β or empty vector. After 12 h, cells were serum-starved for 24 h and then treated with or without phenylephrine (PE; 50 μ M) for various times as indicated in the figure legends. Cells were harvested in lysis buffer and clarified by centrifugation (31). HA-IKK β was immunoprecipitated with an anti-HA mAb (Santa Cruz Biotechnology) for 3 h at 4°C. The activity of the immunocomplex was assayed at 30°C for 30 min in 30 μ l of kinase buffer (31) in the presence of 10 μ M ATP/10 μ Ci [γ -³²P]ATP (10 Ci/mmol) with glutathione *S*-transferase-I κ B α as a substrate (31). The reactions were terminated with Laemmli sample buffer. The proteins were resolved by 13% SDS/PAGE, followed by autoradiography. The phosphorylated proteins were quantitated by a PhosphorImager (Molecular Dynamics). Immunoblot analysis was performed as described (31).

Calculation of Cardiomyocyte Size. Primary cardiomyocytes grown on coverslips were transfected with various expression vectors or empty vector, as described in the legend to Fig. 1. Cells were treated with or without PE (50 μ M) for 4 days. Transfected cells were identified by indirect immunofluorescence analysis as described (30). Cardiomyocytes were visualized by using an Olympus fluorescence microscope (New Hyde Park, NY), which was connected to a Sony video monitor to allow direct computation of the surface area of the cardiomyocytes by using image analysis software (SCION IMAGING; National Institutes of Health, Rockville, MD). Cells from randomly selected fields in three culture plates were examined and the surface area of cells from each group was determined and compared with control.

Results

Stimulation of NF- κ B Activity by Hypertrophic Agonists in Primary Cardiomyocytes. To study the role of NF- κ B in regulating hypertrophic growth of cardiomyocytes, we tested whether NF- κ B is activated during cardiomyocyte hypertrophy.

Primary cardiomyocytes grown on coverslips were serum-starved and treated with PE, which is a pharmacological agonist of the α_1 adrenergic receptor that is able to induce cardiac hypertrophy (28), or endothelin-1 (ET-1) and angiotensin II (AngII), two physiological agonists that can induce cardiac hypertrophy (28). All three agonists stimulated the nuclear translocation of p65, a member of the NF- κ B family, as measured by indirect immunofluorescence assays (Fig. 1A). The nuclear

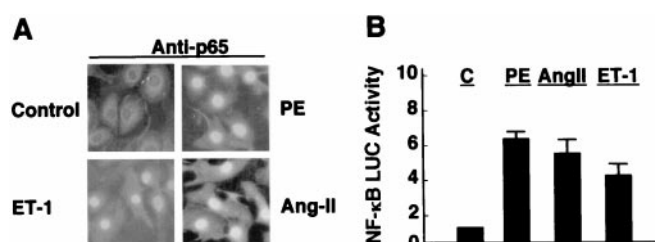


Fig. 1. Stimulation of NF- κ B activity by various hypertrophic agonists in primary cardiomyocytes. (A) Cardiomyocytes grown on coverslips were serum-starved for 24 h before treatment with PE (50 μ M), ET-1 (50 nM), and AngII (50 μ M) for 30 min or left untreated. Cells were fixed, blocked, and stained with anti-p65 mAb followed by donkey anti-mouse FITC (The Jackson Laboratory). The nuclear translocation of p65/RelA was visualized by using an Olympus fluorescence microscope. (B) Primary cardiomyocytes were transfected with the 2 $\times$ NF- κ B reporter plasmid (0.25 μ g/plate) and treated with PE (50 μ M), AngII (50 μ M), and ET-1 (50 nM) for 24 h or left untreated. Cells were harvested, and the Luc activity was determined and normalized to the protein content of each extract. Luc activity expressed by cells transfected with empty vector was given an arbitrary value of 1. The results are presented as the mean $\pm$ SE and represent four individual experiments.

translocation of p65 was detected 15 min after treatment, with kinetics similar to that induced by tumor necrosis factor (TNF) α (data not shown) (7, 31).

To determine whether these hypertrophic agonists are able to stimulate NF- κ B transcriptional activity, primary cardiomyocytes were transfected with the 2 $\times$ NF- κ B-Luc reporter gene and treated with PE, ET-1, and AngII. All three agonists significantly stimulated NF- κ B transcriptional activity (Fig. 1B) in a dose-dependent manner (data not shown), as measured by the activity of the NF- κ B-Luc reporter gene (31). Thus, NF- κ B appears to be a novel target of these hypertrophic agonists in cardiomyocytes.

Activation of NF- κ B in Cardiomyocytes Depends on Degradation of I κ B α Proteins and Activation of the IKK Complex. A key regulatory step in NF- κ B activation is stimulation-induced, ubiquitination-dependent degradation of I κ B proteins by the 26S proteasome (4, 18, 19), a process catalyzed by the IKK complex (6–14). However, NF- κ B also can be activated independently of stimulation-induced degradation of I κ B proteins and IKK activation (34–37). To determine the molecular mechanism of NF- κ B activation during hypertrophic growth of cardiomyocytes, primary cardiomyocytes were cotransfected with the 2 $\times$ NF- κ B-Luc gene, with or without expression vectors encoding the I κ B α (32Ala/36Ala) mutant, which is resistant to stimulation-induced degradation and functions as a supersuppressor of NF- κ B activation (15, 16). Cells were treated with PE, ET-1, and AngII for 24 h or left untreated. Expression of the I κ B α (32Ala/36Ala) mutant completely blocked stimulation of the NF- κ B-Luc activity by PE, ET-1, or AngII (Fig. 2A). Treatment with PE induced degradation of endogenous I κ B α proteins (Fig. 2B) and wild-type HA-I κ B α proteins (data not shown). The protein level of actin was not changed (Fig. 2B). The levels of I κ B α protein were decreased 15 min after stimulation by PE and had recovered at 45 min after stimulation, due to the newly synthesized I κ B α , which is one of the target genes of NF- κ B (1–5). Although the degradation of I κ B α proteins by PE was not as complete as that seen in TNF- α -treated cells, it was significant (33% of total proteins) and reproducible. In contrast, the I κ B α (32Ala/36Ala) mutant was not degraded (data not shown), suggesting the degradation of I κ B α proteins was a phosphorylation-dependent process. These results suggest that stimulation-dependent degradation of I κ B α is required for PE-induced NF- κ B activation.

The IKK complex mediates activation of NF- κ B by various extracellular stimuli, such as TNF- α , IL-1, and γ -radiation (38,

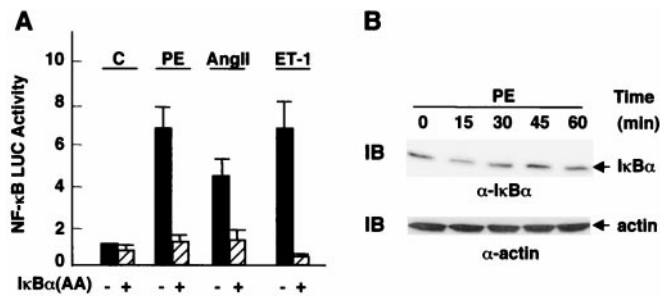


Fig. 2. NF- κ B activation in cardiomyocytes involves I κ B α degradation. (A) Primary cardiomyocytes were transfected with the 2 $\times$ NF- κ B reporter plasmid (0.25 μ g/plate), with or without the supersuppressor I κ B α (32Ala/36Ala) mutant (50 ng/plate). Cells were serum-starved and then treated with PE (50 μ M), AngII (50 μ M), and ET-1 (50 nM) or left untreated. After 24 h, cells were harvested and the Luc activity was determined as described in Fig. 1. The results are presented as the mean $\pm$ SE and represent three individual experiments. (B) Primary cardiomyocytes were serum-starved and treated with PE for the indicated times or left untreated. Cells were harvested, and cell extracts (50 μ g) were analyzed for degradation of I κ B α proteins by immunoblotting with an anti-I κ B α mAb (Santa Cruz Biotechnology). The protein level of endogenous actin also was analyzed to serve as a control.

39). To determine whether activation of NF- κ B in cardiomyocyte hypertrophy is mediated by IKK, primary cardiomyocytes were cotransfected with the 2 $\times$ NF- κ B-Luc gene, with or without expression vectors encoding wild-type HA-IKK β , or empty vector. Cells were treated with PE for 24 h or left untreated. A lower dose of PE (10 μ M) stimulated the NF- κ B-Luc activity $\approx$ 6-fold (Fig. 3A, lane 2). Expression of wild-type IKK β potentiated the stimulation by PE, leading to a 24-fold activation (Fig. 3A, compare lane 3 to lane 2). The potentiation by IKK β was synergistic because expression of IKK β alone only stimulated the NF- κ B-Luc activity $\approx$ 5-fold (Fig. 3A, lane 4). However, IKK β may act in a parallel but separate signaling pathway from the PE-signaling pathway, leading to the stimulation of NF- κ B-Luc activity. Therefore, we examined whether the HA-IKK β (177Ala/181Ala) mutant, which has been previously shown to function as a dominant negative mutant to block NF- κ B activation by many other stimuli, such as TNF- α and IL-1 (8, 31), is able to block the effect of PE on the transcription activity of NF- κ B. Expression of the dominant negative IKK β (177Ala/181Ala) mutant significantly inhibited stimulation of the NF- κ B-Luc activity by PE (Fig. 3B, compare lane 3 to lane 2), suggesting IKK acts downstream of PE.

If the IKK complex mediates the activation of NF- κ B by PE in primary cardiomyocytes, PE should stimulate IKK activity. Indeed, HA-IKK β was transiently activated 10 min after PE treatment (Fig. 3C), with kinetics similar to the activation by TNF- α in other cell types (7, 31). The activation was not a result of increased expression of HA-IKK β , as demonstrated by immunoblotting analysis (Fig. 3C). The activation of endogenous IKK by PE, however, was much weaker because the protein level of IKK in primary cardiomyocytes is extremely low (data not shown). Taken together, these results suggest that in primary cardiomyocytes activation of NF- κ B by hypertrophic agonists such as PE is mediated by IKK activation.

Inhibition of NF- κ B Activation Blocks the Hypertrophic Agonist-Induced Expression of the Embryonic Gene ANF and Cardiomyocyte Enlargement. The experimental results above suggest that activation of NF- κ B may be required for hypertrophic growth of primary cardiomyocytes. To test this scenario, we determined the effect of inhibition of NF- κ B activation on the hypertrophic agonist-induced expression of ANF, which is an important genetic marker of *in vivo* cardiac hypertrophy (28).

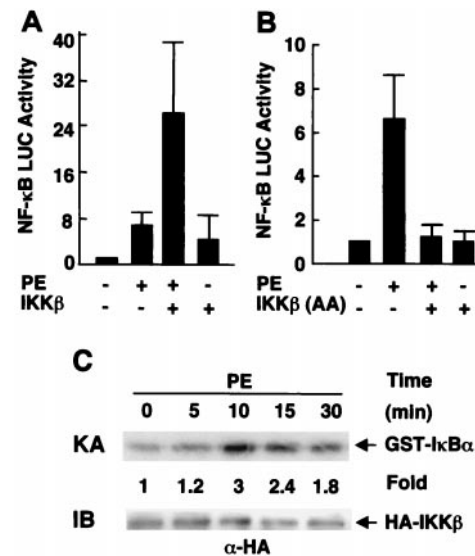


Fig. 3. Activation of NF- κ B by the hypertrophic agonist PE is mediated by the IKK complex. Primary cardiomyocytes were cotransfected with the 2 $\times$ NF- κ B-Luc reporter plasmid (0.25 μ g/plate) and expression vectors encoding HA-IKK β (A) (100 ng/plate) or the dominant negative HA-IKK β (177Ala/181Ala) (10 ng/plate), or empty vector, as indicated. Cells were treated with 10 μ M PE or left untreated. After 24 h, cells were harvested, and the Luc activity was determined as described for Fig. 1. The results are presented as the mean $\pm$ SE and represent five individual experiments. (C) Primary cardiomyocytes were transfected with the expression vector encoding HA-IKK β (3 μ g/plate) or empty vector. After 40 h, cells were treated with PE (50 μ M) for various times, or left untreated, as indicated. HA-IKK β was immunoprecipitated, and its activity was determined by immunocomplex kinase assays with glutathione S-transferase-I κ B α as a substrate, as described (31). Substrate phosphorylation was quantitated with a PhosphorImager. Fold stimulation is indicated. (Lower) An aliquot of each lysate was analyzed for its content of HA-IKK β by immunoblotting, as described (31). ns, nonspecific.

Primary cardiomyocytes were cotransfected with the ANF-Luc gene, along with the expression vector encoding the I κ B α (32Ala/36Ala) mutant, or empty vector. Expression of the HA-I κ B α (32Ala/36Ala) mutant significantly inhibited ANF-promoter activity induced by PE, ET-1, and AngII (Fig. 4A), in a dose-dependent manner (data not shown). Additionally, PE-induced activation of ANF-Luc reporter gene was potentiated by expression of wild-type IKK β (Fig. 4B, compare lane 3 to lane 2) but blocked by expression of the HA-IKK β (177Ala/181Ala) mutant (Fig. 4B, compare lane 5 to lane 2). These results suggest that NF- κ B activation may be necessary for expression of ANF during cardiomyocyte hypertrophy.

Another important feature of cardiomyocyte hypertrophy is the morphological changes of cardiomyocytes, such as enlargement in cell size (27). To determine whether inhibition of NF- κ B activation affects the hypertrophic morphological changes of primary cardiomyocytes, cells were transfected with the HA-I κ B α (32Ala/36Ala) mutant or empty vector. After treatment with PE for 4 days, the transfected cells were identified by indirect immunofluorescence analysis with anti-HA antibody. Measurement of the cell surface area showed that treatment with PE resulted in a 100% increase in cell surface area (Fig. 4C, compare lane 2 to lane 1). Expression of the I κ B α (32Ala/36Ala) mutant inhibited the effect of PE on cell enlargement (Fig. 4C, compare lane 3 to lane 2) but had no effect on the surface area of nonstimulated cells (Fig. 4C, compare lane 4 to lane 1). To further verify this observation, cardiomyocytes were infected with Ad/I κ B α (32Ala/36Ala) dominant negative mutant or control Ad/GFP. The infection efficiency was 100% in cardiomyocytes (Fig. 4D, Ad/GFP). Consistently, treatment with PE

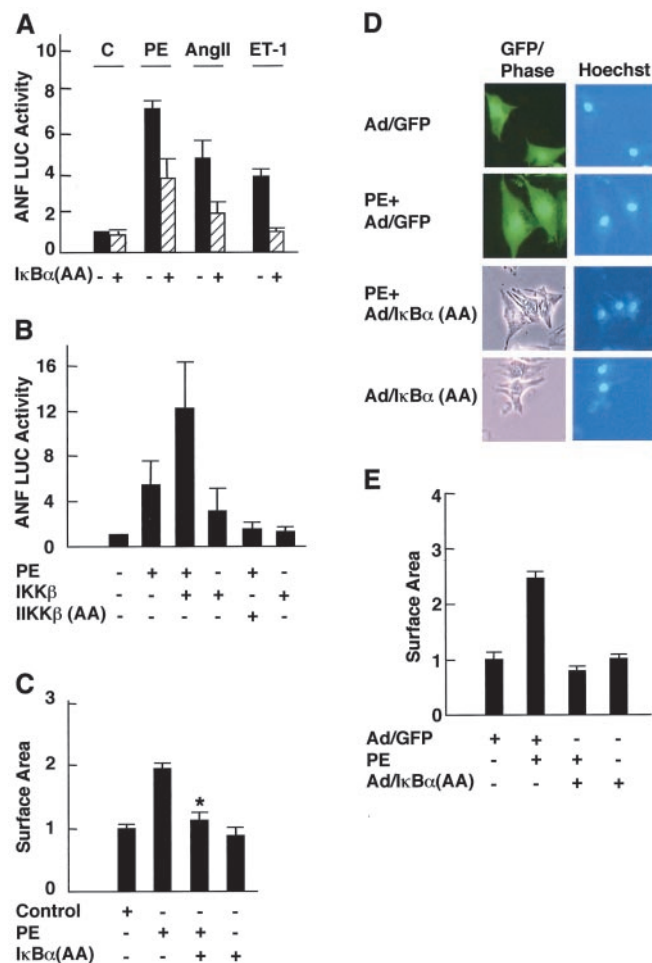


Fig. 4. Inhibition of NF- κ B activation blocks the hypertrophic effects of PE, ET-1, and AngII on cardiomyocytes. Primary cardiomyocytes were transfected with the ANF-Luc reporter plasmid (1 μ g/plate), with or without (A) the supersuppressor HA-I κ B α (32Ala/36Ala) mutant (50 ng/plate), (B) wild-type HA-IKK β (10 ng/plate), or the dominant negative HA-IKK β (177Ala/181Ala) mutant (10 ng/plate). Cells were treated with PE (50 μ M), ET-1 (50 nM), and AngII (50 μ M) or left untreated. After 24 h, cells were harvested and the Luc activity was determined as described in Fig. 1. The results are presented as the mean $\pm$ SE and represent four individual experiments. (C) Primary cardiomyocytes grown on coverslips were transfected with the expression vector encoding the dominant negative HA-I κ B α (32Ala/36Ala) mutant and treated with PE (50 μ M). After 4 days, transfected cells were identified by indirect immunofluorescence with anti-HA mAb (Covance, Richmond, CA). Cells were visualized by using an Olympus fluorescence microscope, and the surface area of cells were measured by using SCION IMAGING software. *, $P < 0.02$. (D and E) Primary cardiomyocytes grown on coverslips were infected with Ads encoding the dominant negative I κ B α (32Ala/36Ala) mutant or GFP at a multiplicity of infection of 100. After 16 h, cells were treated with or without PE (50 μ M). Cell surface areas were measured 4 days later as described in C. The data were from three individual experiments with $n = 60$ cells, presented as mean $\pm$ SE. *, $P < 0.02$.

induced $\approx 100\%$ enlargement in cell surface area in Ad/GFP-infected cells (Fig. 4D and E), which was similar to its effects on noninfected myocytes (Fig. 4C). However, the PE effect on cell enlargement was completely abolished in cells infected with Ad/I κ B α (32Ala/36Ala) mutant (Fig. 4D and E). Neither Ad/I κ B α (32Ala/36Ala) mutant nor Ad/GFP had detectable effect on cell enlargement (Fig. 4D and E). Taken together, these results suggest that activation of NF- κ B may be necessary for the enlargement of cardiomyocytes during hypertrophic growth.

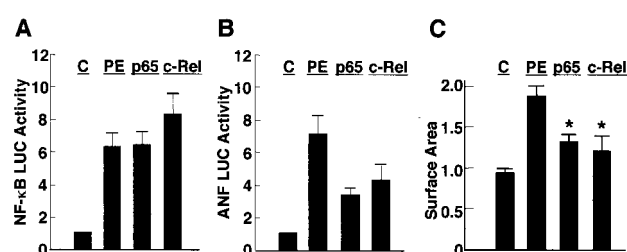


Fig. 5. Overexpression of members of the NF- κ B family, p65 and c-Rel, induces ANF expression and cell enlargement in cardiomyocytes. Primary cardiomyocytes were cotransfected with the 2 $\times$ NF- κ B-Luc reporter plasmid (A) (0.25 μ g/plate) or the ANF-Luc reporter plasmid (B) (1 μ g/plate) with the expression vector encoding p65 or c-Rel (50 ng each), or empty vector, as indicated. After 24 h, cells were harvested and the Luc activity was determined as described for Fig. 1. The results are presented as the mean $\pm$ SE and represent five individual experiments. (C) Primary cardiomyocytes grown on coverslips were transfected with the expression vector encoding HA-p65 or c-Rel (2 μ g each). After 4 days, transfected cells were identified by indirect immunofluorescence with anti-HA mAb (Covance) or anti-c-Rel mAb, respectively (Santa Cruz Biotechnology). Cells were visualized by using an Olympus fluorescence microscope, and the surface area of cells were measured by using SCION IMAGING software. The results are presented as mean $\pm$ SE and represent three individual experiments with $n = 30$. *, $P < 0.01$.

Activation of NF- κ B Itself Induces Cardiomyocyte Hypertrophy. Previous studies have shown that overexpression of NF- κ B allows it to overcome the inhibition by I κ B proteins (40). Thus, we tested whether overexpression of p65 and c-Rel, members of the NF- κ B family, can induce hypertrophic growth of cardiomyocytes.

Primary cardiomyocytes were cotransfected with the ANF-Luc reporter gene or the 2 $\times$ NF- κ B-Luc reporter gene, with or without expression vectors encoding HA-p65, c-Rel, or empty vector. Expression of HA-p65 or c-Rel stimulated the activity of the 2 $\times$ NF- κ B-Luc reporter gene (Fig. 5A). Expression of HA-p65 or c-Rel also stimulated the activity of the ANF-Luc reporter gene (Fig. 5B). The stimulation by p65 or c-Rel was significant, although it was weaker than the effect of PE (Fig. 5B, compare lane 3 and 4 to lane 2), consistent with the notion that several signaling pathways function downstream of PE in cardiomyocyte hypertrophy (30).

To determine the effect of NF- κ B overexpression on hypertrophic morphology changes of cardiomyocytes, cells were transiently transfected with expression vectors encoding HA-p65, c-Rel, or empty vector. The transfected cells were identified by indirect immunofluorescence analysis. Measurement of the surface area of cells revealed that overexpression of HA-p65 or c-Rel caused a 30–40% increase in cell size compared to that of control (Fig. 5C, compare lanes 3 and 4 to lane 1). The effect of overexpression of HA-p65 or c-Rel on cell size was significant ($P < 0.01$), although it was weaker than the effect of PE, which caused a 100% increase in cell size (Fig. 5C, compare lane 2 to lane 1). These results indicate that NF- κ B itself is able, at least in part, to induce cardiomyocyte hypertrophy and suggest that signaling by NF- κ B may be critical for hypertrophic growth of primary cardiomyocytes.

Discussion

NF- κ B is known to be involved in inflammation, the immune response, viral infection, and regulation of programmed cell death (39). In this report, we show that activation of NF- κ B is also required for hypertrophic growth of rat primary neonatal ventricular cardiomyocytes. This conclusion is based on several lines of evidence. First, several hypertrophic agonists, PE, ET-1, and AngII, stimulate the nuclear translocation of NF- κ B and its transcriptional activity in primary cardiomyocytes (Fig. 1A and B). The activation of NF- κ B depends on I κ B α degradation

because it is inhibited by expression of the supersuppressor I κ B α (32Ala/36Ala) mutant (Fig. 2A), and PE induces degradation of endogenous I κ B α proteins (Fig. 2B). Furthermore, NF- κ B activation by PE is potentiated by expression of wild-type IKK β (Fig. 3A) but inhibited by the dominant negative IKK β (177Ala/181Ala) mutant (Fig. 3B). Additionally, PE also stimulates HA-IKK β activity (Fig. 3C). Second, the hypertrophic agonist-induced expression of ANF is inhibited by expression of the I κ B α (32Ala/36Ala) mutant (Fig. 4A), or the dominant negative IKK β (177Ala/181Ala) mutant (Fig. 4B), but is potentiated by wild-type IKK β (Fig. 4B). The enlargement of cardiomyocytes induced by PE also is blocked by expression of the I κ B α (32Ala/36Ala) mutant (Fig. 4C–E). Finally, overexpression of HA-p65 or c-Rel, members of the NF- κ B family, is able to induce hypertrophic growth of cardiomyocytes, as evident by the induction of ANF expression and enlargement of cardiomyocytes (Fig. 5B and C). We show direct evidence that activation of NF- κ B is required for hypertrophic growth of terminally differentiated cardiomyocytes.

The kinetics of NF- κ B translocation, I κ B α degradation, and IKK β activation by the hypertrophic agonists PE, ET-1, and AngII are similar to that induced by TNF- α in other cell types (7, 31). Furthermore, the dominant negative mutants of MEKK1 and NIK, two of the upstream kinases of the IKK complex, also blocked activation of NF- κ B by PE (data not shown), suggesting these upstream kinases also may be involved in NF- κ B activation by these agonists. Thus, the hypertrophic agonists may share a common signaling pathway that is used by TNF- α .

The detailed mechanism by which NF- κ B regulates hypertrophic growth of cardiomyocytes has yet to be determined. As a transcription factor, NF- κ B may regulate ANF expression directly. Although analysis of the promoter region of ANF did not identify a classic NF- κ B-binding site (data not shown), NF- κ B may bind to atypical binding sites to stimulate ANF expression (41). Another possibility is that NF- κ B may regulate ANF expression indirectly, through its interaction with yet-to-be identified cardiomyocyte specific cofactors (30). Because NF- κ B can induce synthesis of contractile proteins such as myosin heavy

chain in terminally differentiated smooth muscle cells (26), it is possible that in cardiomyocytes NF- κ B also may regulate the expression of myosin heavy chain and myosin light chain proteins, whose accumulation is involved in cell enlargement (28). Additionally, the fact that NF- κ B can be activated by the Rho family proteins Rac and Cdc42 also supports the role of NF- κ B in enlargement of cardiomyocytes because Rac and Cdc42 are involved in regulating cytoskeleton organization (42). It would be of interest to examine these possibilities.

Hypertrophic growth of cardiomyocytes is a complex adaptive response (43). It has been reported that many cellular signaling pathways, such as the mitogen-activated protein kinases (ERK, JNK, and p38) (30, 44–48), gp130 (36), and protein phosphatase calcineurin (49, 50), are involved in cardiomyocyte hypertrophy. It is not clear and sometimes controversial regarding which signaling pathway is critical for cardiomyocyte hypertrophy (51–53). Given the fact that the transmission of extracellular signals are mediated by a signaling network, rather than a single signaling pathway, and that many factors are involved in hypertrophic growth of cardiomyocytes, the growth of cardiomyocytes is likely the result of cooperation among many signaling pathways (54, 55). For instance, recent studies showed that activation of the p38 pathway may contribute to NF- κ B transcriptional activity (55–58), most likely through enhancing expression of NF- κ B target genes (56, 58). Future studies are needed to explore the interplay between the NF- κ B and other signaling pathways and to elucidate the *in vivo* role of NF- κ B activation in cardiomyocyte hypertrophy. The identification of NF- κ B as a novel player in regulating cardiomyocyte hypertrophy, however, provides a potential target for prevention and treatment of heart disease.

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